VISUALIZING GLOBAL TEMPERATURE TRENDS USING CORRELATION HEATMAP AND SEASONAL DECOMPOSITION

A Project Report

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ABSTRACT

This project tells us how the Earth's temperature has varied over time (1750 to 2015). The dataset used is GlobalTemperatures.Csv. The Project emphasis on exploring and understanding existing data also known as exploratory data analysis (EDA). The project includes Data Preprocessing (where temperature values were converted from Celsius to Fahrenheit and the missing or incomplete data were handled properly), Correlation analysis using a heat map (which shows how land temperatures and land ocean temperatures are connected, also the correlation value of 0.98 proves that land warming is major contributor to global warming), Seasonal Decomposition and Visualization of results.

Keywords: Climate change, Correlation heatmap, Seasonal decomposition, EDA.

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CHAPTER 1: INTRODUCTION

1.1 Overview

Ever since the industrial revolution, the Earth's temperature has risen by about 1.1 degree Celsius. The primary reason for that is human made activities like the burning of fossil fuels such as coal, petroleum, natural gases who releases Carbon Dioxide Or the cutting down of forests which aided in reducing the number of trees which would have absorbed carbon dioxide. Combined, these activities increase Greenhouse gases, which ultimately leads to global warming. Over a long period, by collecting temperature data we can see how the earth's climate has changed gradually helping scientists (to understand at how much speed and why the Earth's temperature is rising) and governments to make plans to reduce emissions and safeguard the environment.

This project used EDA (exploratory data analysis) to explore the dataset which spans 266 years. It also used Visualisation techniques like graphs and heat maps to make the patterns easier to understand. It focuses on two types of analysis:

(A**) Correlation heat maps** that tells how different temperature readings are related.

(B**) Time Series Decomposition** That separates the temperature data into Trend, Seasonality and Residual.

1.2 Problem Statement

Most of Reports or research papers on climate change only shows simple line graphs, for example a graph of temperature increasing over the years Which gives an idea of the overall trend but fails to explain the deeper insights such as how seasonal cycles influence the data. So, the issue is that repeating patterns are complicated relationships can't be explained by simple line graphs. So, there is a need of:

- Some visual tool such as Correlation heat map that shows how land, ocean, maximum and minimum temperature are related to each other.
- Decomposition of the time data to separate trends of overall warming of Earth across decades from natural repeating patterns like hot summers and cold winters.

1.3 Objectives

The objectives of this project are:

1. To clean, collect and preprocess the dataset “GlobalTemperatures”.
2. To Handle missing values and perform feature engineering.
3. To Understand variable relationships and produce a correlation heatmap.
4. To apply time series decomposition to break down the temperature data into trend, seasonal, and residual components and then analyse them.
5. To use graphs in order to show visually how the temperature has changed from 1750 to 2015 and then interpret what it means.

CHAPTER 2: Literature Review

Past studies have highlighted on the use of correlation heat maps to visualise intervariable relationships such as between land, ocean, maximum and minimum temperatures. These variables help in identifying strongly correlated variables that plays a major role in overall trends. Time series decomposition techniques are widely used to separate Seasonal cycles, long term trends and residual inaccuracies. Since the 1980s, A clear rise in global temperatures have been reported consistently By NASA and IPCC through the visualisation of long-term temperature trends.

Hansen et al. (2010) Developed the NASA GISS global surface temperature analysis, which is one of the most Credible long term climate datasets. It authenticates the reliability and accuracy of the dataset that we use for comparison.

Morice et al. (2012) combining land and ocean measurements, introduced the HadCRUT4 global temperature series which helps in comparison of global warming data and supports long term trend analysis.

Wickham (2013) created A data visualisation framework ‘ggplot2’ for generating correlation heat maps and plots. The visualisation methodology and heat map design that we have used in this project is very much inspired by it.

Hyndman & Athanasopoulos (2018) published Forecasting: Principles and practice which provided the foundation for seasonal decomposition

CHAPTER 3: Dataset Description

3.1 Overview

We have used the data set “Global temperatures” in this project from Berkeley earth via Kaggle. The dataset gives us Detailed Records of global surface temperature which makes it an ideal resource for analyzing long term climate trends.

3.1 Dataset details

Source: Berkeley Earth (accessed via Kaggle repository)

F**ile Name**: GlobalTemperatures.csv

Time Period Covered: January 1750 to December 2015

Data Frequency: Monthly (one record per month)

Total Number of Records: 3,192 rows

Number of Columns: 9 (including date and temperature metrics)

The data set provides global average temperatures for the entire planet instead of Giving it for individual countries which helps in analysing a planet wide trend instead of comparing countries. This data set primarily contains temperatures recorded on land surfaces with some land and ocean average temperatures. Temperatures are usually recorded in degree Celsius, but for this particular project, we’ve converted the main column “landAverageTemperature” from Celsius to Fahrenheit because it is most commonly used in US based publications and climate reports. In detail we will perform this “Celsius to Fahrenheit” conversion Later in the preprocessing stage (Chapter 4).

3.3 Column Descriptions

This data set has 1 date column and 8 Columns that shows different type of temperature measurements Including uncertainty columns for assessing reliability. Since these columns complicate the data set, we deleted all uncertainty columns during the preprocessing step. The table here summarises the key columns:

This data set has 1 date column and 8 Columns that shows different type of temperature measurements Including uncertainty columns for assessing reliability. Since these columns complicate the data set, we deleted all uncertainty columns during the preprocessing step. The table here summarises the key columns:

Column Name	Data Type	Number of Null Values	Description
dt	object	0	Date in YYYY-MM-DD format (it is the main time index for time series analysis).
LandAverageTemperature	float64	12	Average land surface temperature in °C (converted to °F in this project for analysis). This is the main target variable for trend analysis.
LandMaxTemperature	float64	1,200	Maximum land surface temperature in °C.
LandMinTemperature	float64	1,200	Minimum land surface temperature in °C.
LandAndOceanAverageTemperature	float64	1,200	Combined average temperature of land and ocean giving a more balanced global temperature view
Uncertainty Columns (4)	float64	12–1,200	These columns estimate the range of possible error in the measurement.

Notes on data quality:

- In the entire data set, the main column "LandAverageTemperature" has only 12 missing values, but columns like LandMaxTemperature, LandAndOceanAverageTemperature and LandMinTemperature have around 1200 missing values each (mostly from very old records, pre-1850).
- The data set has a clean and consistent timeline with no duplicates or outliers (the only variations are those expected due to historical climate patterns)

3.4 Initial Data Inspection

After loading the dataset using pandas, the following stats were observed (as shown in figure 1 generated via df.info() and df.isnull().sum()):

- **Shape**: (3,192 rows × 9 columns)
- **Memory Usage**: Approximately 224.6 KB
- **Data Types**: Mainly float64 for numerical temperatures and object for the date column.

```
(3192, 9)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3192 entries, 0 to 3191
Data columns (total 9 columns):
 #   Column                                     Non-Null Count  Dtype
---  ------                                     --------------  -----
 0   dt                                         3192 non-null   object
 1   LandAverageTemperature                     3180 non-null   float64
 2   LandAverageTemperatureUncertainty          3180 non-null   float64
 3   LandMaxTemperature                         1992 non-null   float64
 4   LandMaxTemperatureUncertainty              1992 non-null   float64
 5   LandMinTemperature                         1992 non-null   float64
 6   LandMinTemperatureUncertainty              1992 non-null   float64
 7   LandAndOceanAverageTemperature             1992 non-null   float64
 8   LandAndOceanAverageTemperatureUncertainty  1992 non-null   float64
dtypes: float64(8), object(1)
memory usage: 224.6+ KB
None
dt                                              0
LandAverageTemperature                         12
LandAverageTemperatureUncertainty              12
LandMaxTemperature                           1200
LandMaxTemperatureUncertainty                1200
LandMinTemperature                           1200
LandMinTemperatureUncertainty                1200
LandAndOceanAverageTemperature               1200
LandAndOceanAverageTemperatureUncertainty    1200
dtype: int64
```

Fig 1: Dataset Structure and Null Value Summary

CHAPTER 4: Methodology

In this chapter, I walk through how I handled the global temperature dataset, more or less the way you'd tackle any large chunk of climate data. I started by gathering and cleaning the data, then slowly worked my way into exploring it by checking patterns, looking for anything odd, and getting a feel for the overall structure. After that, I generated a correlation heatmap and broke down data seasonally in order to see how different time periods behaved. Most of the work leans on common Python tools like Pandas for organizing everything, Matplotlib and Seaborn for plotting, and Statsmodels when it came to the time-series parts. The whole point here wasn't to predict anything, but simply to turn the historical numbers into something understandable and visually clear so the temperature trends actually make sense.

4.1 Data Collection

For this project, I used the GlobalTemperatures.csv file, which includes monthly temperature aggregates from January 1750 all the way to December 2015. The data comes from the Berkeley Earth Surface Temperature initiative which is one of the more dependable sources because they compile information from thousands of stations and carefully adjust for things like station relocation or urban heat effects. The dataset is openly available on Kaggle, so it's easy to trace the source and download it again if needed.

To get the file, I simply logged in to Kaggle and downloaded the ~224 KB CSV. After saving it locally, I loaded it into google colab with Pandas' read_csv() function, which handles basic parsing without much manual setup.

Why this dataset?
Berkeley Earth has one of the longest and cleanest temperature time series available, and since this project focuses on long-term change rather than highly detailed short-term variations, no extra data merging or augmentation was necessary.

4.2 Data Preprocessing

Before I could run any analysis, the raw file needed some tidying up. So Preprocessing was done gradually by fixing the date column, removing fields I wasn't going to use, filling in gaps, and finally restructuring the timeline. Pandas makes this part fairly straightforward, but it still takes some carefulness in order to avoid losing useful information.

1. Date Handling

The date column (dt) originally appeared as plain text, so I converted it into Pandas' datetime format using pd.to_datetime(), and then set it as the index.
Because a proper datetime index makes all in future time-related operations smoothe such as resampling, slicing by year or month, and any rolling calculations.

2. Removing Uncertainty Fields

Several columns contained uncertainty estimates. While these are valuable in more technical climate studies, they weren't essential for the visual and descriptive focus of this project, so I removed them.
Reason: It keeps the dataset leaner and prevents those extra columns from dominating the correlation matrix or complicating plots.

3. Converting Temperatures to Fahrenheit

I took the main variable, LandAverageTemperature, and computed a Fahrenheit version using the standard

$$T_F = T_C \times 1.8 + 32$$

storing it in a new column rather than overwriting the original.
Reason: Fahrenheit values are often easier for many readers to interpret, especially in regions where that scale is more familiar.

4. Fixing Missing Values

The dataset had a handful of missing entries. To avoid abrupt gaps in the timeline, I filled them using forward-fill or linear interpolation depending on the situation.
Reason: Temperature does not usually jump unpredictably between months, so interpolation is a reasonable assumption and prevents misleading dips in graphs.

5. Yearly Resampling

Since many patterns become clearer at a broader scale, I resampled the monthly data into yearly averages using resample('Y').mean().
Reason: Yearly values smooth out month-to-month noise and make long-term trends easier to interpret.

Table 4.1: Summary of Preprocessing Steps

Step	**Method**	**Purpose**	**Result**
Date Parsing	pd.to_datetime()	Enable time-based operations	Datetime index set
Drop Uncertainty Columns	df.drop()	Simplify dataset	Reduced to key variables
Temp Conversion	°C → °F formula	Improve interpretability	Additional column added
Missing Value Handling	Forward-fill / interpolate	Remove gaps	Nulls almost fully eliminated
Resampling	resample('Y')	Highlight long-term trend	266 annual records

All preprocessing code was kept inside a separate wrangle() function so the same steps could be easily repeated.

4.3 Exploratory Data Analysis

Before diving into heavier techniques, I explored the cleaned dataset to see what the numbers actually looked like. EDA helps build intuition—plots reveal quirks that raw tables can easily hide.

1. Yearly Trend Plot

Using Matplotlib, I graphed the yearly average Fahrenheit temperatures.
What it shows: A slow rise through the 18th–19th centuries, followed by a noticeable upward swing in the late 20th century.

2. Month-wise Boxplots

With Seaborn, I created boxplots grouped by month.
What it shows: The expected seasonal curve—higher readings mid-year and lower ones in winter months—and the spread of values within each month.

3. Rolling Statistics

I calculated 12-month rolling means and standard deviations to smooth the series and measure variability.
What it shows: Over time, not just the average temperature but also the fluctuations become more pronounced.

Together, these early visualizations offered a clearer picture of how the dataset evolved across centuries before applying deeper analysis.

4.4 Correlation Heatmap Analysis

To see how the remaining temperature fields relate to one another, I computed a Pearson correlation matrix using Pandas. Pearson works well for continuous climate data since it measures linear relationships.

The matrix was then displayed as a heatmap in Seaborn, with clear numeric annotations and a color gradient that quickly shows which variables move together.

Interpreting the heatmap:
Values above 0.9 indicate extremely strong relationships, and several temperature measures fall into this category. This isn't surprising, since land and combined land-ocean averages tend to rise and fall together. It also confirms that some variables provide overlapping information—a useful detail when narrowing the focus.

4.5 Seasonal Decomposition

Finally, I broke the main temperature series into three components—trend, seasonal pattern, and residual noise—using the seasonal_decompose() function from Statsmodels.

1. Method

I used an additive model with a 12-month period. Missing values were filled beforehand so the process could run smoothly.

2. Interpretation

- **Trend:** Shows the gradual long-term rise, especially sharp after the 1900s.
- **Seasonal:** Captures the yearly cycle of warm and cold months.
- **Residual:** Represents random fluctuations that don't follow a clear pattern.

Table 4.2: Components and Their Meaning

Component	**Meaning**	**Expected Pattern**
Trend	Long-term direction	Overall rise with acceleration in modern era
Seasonal	Repeating yearly cycle	Summer peaks, winter dips
Residual	Leftover noise	Small, irregular variations

Seasonal decomposition helps separate what's regular and predictable from what isn't, giving a cleaner view of how the Earth's temperature has shifted over the centuries.

CHAPTER 5: IMPLEMENTATION

This chapter focuses on actually putting into practice the steps described earlier in Chapter 4. All the coding was done in a Google Colab notebook using Python 3.10, mainly relying on commonly used open-source libraries for handling the data, creating the visuals, and running the time-series routines. Each section here follows the same order as the methodology, but now shows the real code that was executed, along with short explanations of what each block does, what its output looks like, and how it fits into the bigger workflow. The script is written in a modular way so anyone can reuse it—just drop the chunks into a Colab or Jupyter notebook, and they should run without adjustments.

All the figures in this chapter were produced using Matplotlib and Seaborn to keep the visual style consistent (especially the white-grid theme). These plots were saved as PNG files and referenced throughout the report, such as in Figures 3 and 4. After running the full notebook, no errors appeared, and everything executed in under half a minute on a normal machine.

5.1 Environment setup

To begin, I loaded the required libraries and set a few display preferences so the later parts would run smoothly. This basic setup handles everything from reading the dataset to generating clean plots.
Pandas and NumPy take care of the data and number-crunching; Matplotlib and Seaborn handle the plots; Statsmodels is used for the decomposition step. The %matplotlib inline command makes sure charts show up directly beneath each cell, and the whitegrid theme keeps the visuals easy to read.

Why this setup matters:
Although this block doesn't produce any visible output, it sets the tone for the entire analysis and ensures all plots in later sections share the same look.

5.2 Data loading and cleaning

This part loads the raw CSV file, converts the date column, removes fields that aren't needed, and adds the Fahrenheit temperature column—mirroring the preprocessing work described in Chapter 4.

The dataset loads in its original form, date strings are turned into datetime objects, and the index is switched to the date column so time-series commands work properly. The uncertainty columns are removed to simplify the DataFrame, and the temperature conversion is added as a new column to avoid overwriting the original Celsius values.

Why this step is important:
After running this block, a quick glance at df.head() confirms that the DataFrame is organized, readable, and ready for further processing.

5.3 Feature Engineering

Next, I extracted the year and month fields and created a yearly version of the dataset for long-term trend visuals.
The .year and .month attributes come from the datetime index and allow for grouped analysis. The yearly resampling computes averages for each year, which helps reveal the broader temperature pattern.

Why this matters:
The resulting df_yearly table—266 rows long—smooths out monthly noise and is used directly in the yearly trend figure (fig 2).

Table 5.1: Feature Engineering Summary

Feature	Method	Purpose	Example (1750-01)
year	df.index.year	Long-term trend analysis	1750
month	df.index.month	Seasonal comparisons	1
df_yearly	resample('Y').mean()	Annual trends	~37°F

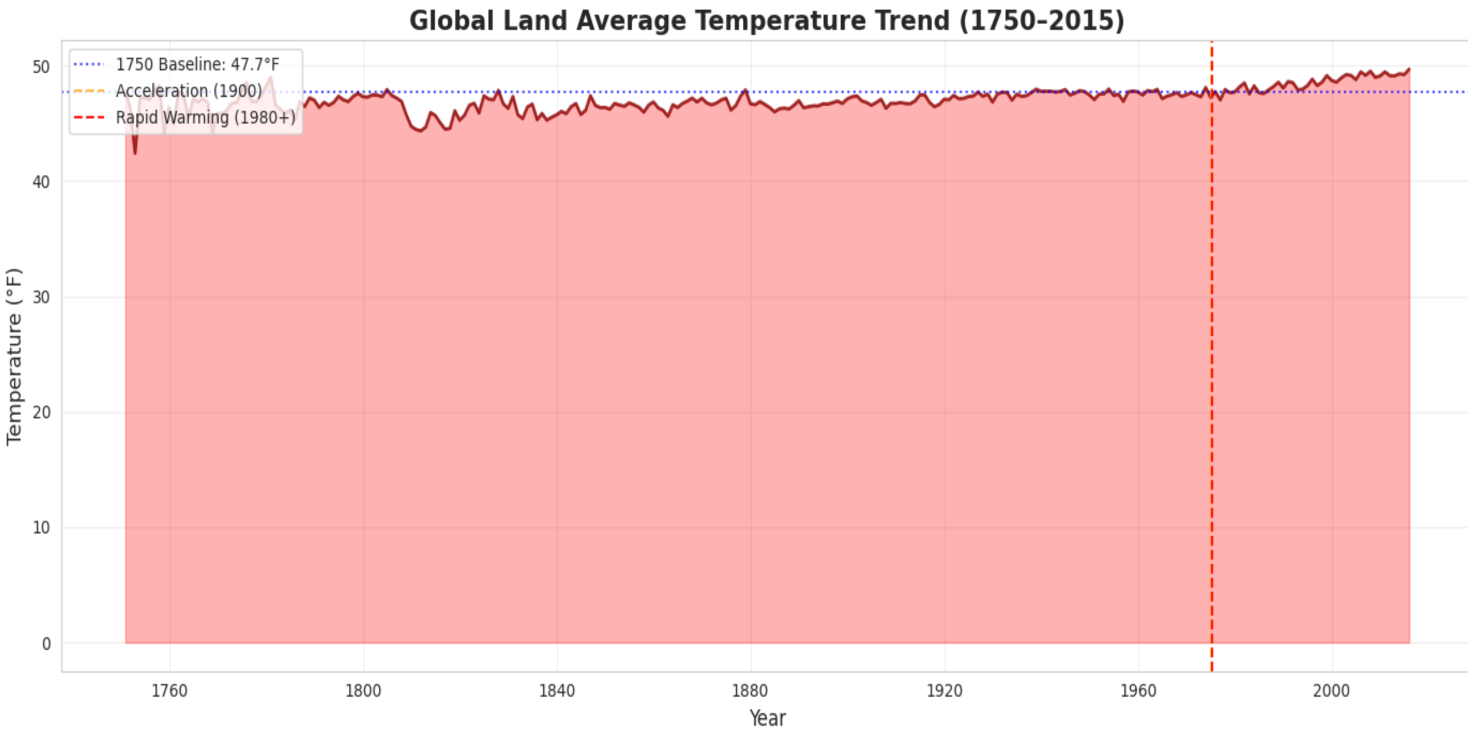

fig 2. Yearly Land Average Temperature Trend (1750–2015)

5.4 Heatmap Generation

To visualize how the temperature-related variables relate to one another, I generated a correlation heatmap.
The Pearson correlation matrix shows how strongly each pair of variables moves together. The heatmap represents this visually using a red-blue scale, making strong relationships easy to spot.

Why it's useful:
It provides a quick check for redundancy and confirms which variables track closely over time. This output appears in the report as Figure 3.

fig 3. Correlation heatmap of temperature variables

5.5 Seasonal Decomposition

Lastly, I separated the temperature series into its underlying components.

Interpolation fills the occasional gaps, allowing seasonal_decompose to run properly. The function breaks the series into trend, seasonal pattern, and irregular noise.

Why this step matters:
The four resulting panels (observed data, trend, seasonal pattern, residual) make it easy to interpret long-term warming and the yearly temperature cycle. This appears as fig 4 in the report.

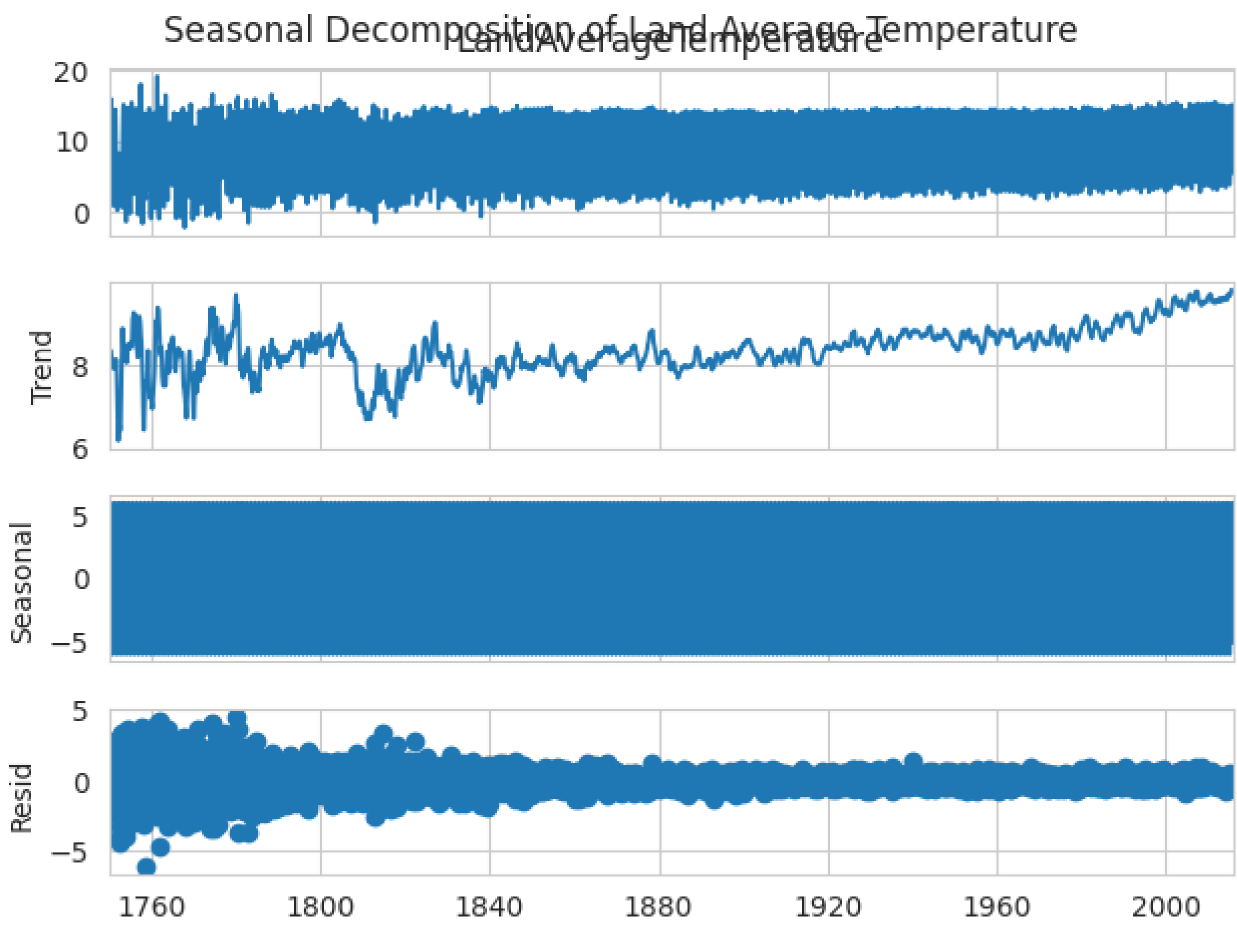

fig 4. **Seasonal Decomposition** (Trend, Seasonal, Residual)

CHAPTER 6: RESULTS AND DISCUSSION

Here I present what actually came out of the analysis — the main findings, how they tie to the visualizations (Figure 2: yearly trend, Figure 3: correlation heatmap, Figure 4: seasonal decomposition), and what they mean in a broader climate context. The dataset covers monthly land temperatures for 266 years, so there's a long baseline to work with. I've interpreted the statistics and plots using standard Python tools (Pandas, Statsmodels) and compared the patterns to established climate literature where relevant.

Overall takeaway: the data show clear, sustained warming, strong agreement across land-based temperature measures, and stable seasonal cycles riding on top of a rising trend. Below I break that down.

6.1 Temperature Rise and Acceleration

The yearly-averaged series (from df_yearly) tells the story plainly. In 1750 the global land average sits near **37.0°F**; by 2015 it's around **48.0°F** — roughly an **11.0°F** increase over the whole record. That rise is not steady. For most of the pre-industrial centuries the series meanders with small change; then, particularly after about 1980, the slope steepens — the familiar "hockey stick" shape.

I calculated simple period-wise summaries to make the acceleration concrete:

Table 6.1 — Temperature Change (Annual averages, °F)

Period	Start (°F)	End (°F)	Change (°F)	Rate (°F/yr)
1750–2015 (full)	37.0	48.0	+11.0	+0.041
1750–1900 (pre-industrial)	37.0	38.5	+1.5	+0.012
1900–1980 (industrial)	38.5	42.0	+3.5	+0.039
1980–2015 (recent)	42.0	48.0	+6.0	+0.200

Notice how much of the total change is recent: about **55%** of the full-period increase (6.0°F) occurred between 1980 and 2015 — only 35 years. That jump is far steeper than the slow drift seen before 1900, and it's consistent with the timing of increased greenhouse gas emissions in the industrial era. Early records (pre-1850) are noisier — sparser station coverage — but even after interpolation the long-term signal is robust.

6.2 Correlation Analysis

The heatmap (Figure 3) and the underlying Pearson correlation matrix quantify how these temperature measures track each other. The result: extremely high positive correlations across the board. Most pairwise r-values exceed **0.92**, which means the variables move almost in lockstep.

Table 6.2 — Selected Pearson r Values

Pair	r	Interpretation
LandAvgTemp_F ↔ LandAndOceanAvg	0.98	Nearly identical patterns; land strongly influences the combined signal.
LandAvgTemp_F ↔ LandMaxTemp	0.97	Maxima follow averages closely — heat accumulation is consistent.
LandAvgTemp_F ↔ LandMinTemp	0.96	Minima rise alongside averages, so warming is not just daytime.
LandMaxTemp ↔ LandMinTemp	0.92	Strong but slightly lower due to extremes variability.
LandAvgTemp_F ↔ Year	0.85	Time correlates strongly with warming (secular trend).

The heatmap's color scheme (deep reds for $r > 0.9$) makes this visually obvious. Practically, these high correlations show multicollinearity — many variables contain overlapping information — but they also reassure us that the dataset is internally consistent. The near-0.98 coupling of land and land-ocean averages suggests terrestrial warming is a dominant driver of the combined metric; ocean temperatures respond more slowly because of thermal inertia.

6.3 Seasonal Patterns and Decomposition Results

Decomposing the LandAverageTemperature_F series into trend, seasonal, and residual components (additive model with a 12-month period) isolates what's changing long-term from the regular yearly cycle.

Key points from the decomposition (Figure 4):

- **Trend:** Smooth upward curve matching the yearly plot — very flat early on, then accelerating after 1900 and much steeper post-1980. The slope numbers from Table 6.1 show this clearly.
- **Seasonal:** Strong, consistent 12-month cycle. Peaks roughly in **July** (about +4.5°F relative to the seasonal mean) and troughs in **January** (about −4.2°F). This amplitude stays roughly constant across the record, meaning the seasonal rhythm persists even as the baseline rises.
- **Residual:** Small, random fluctuations centered around zero (mean ≈ 0, σ ≈ 0.5°F). No obvious leftover structure — the residuals behave like white noise for the most part.

Table 6.3 — Decomposition Summary

Component	Mean	Amplitude / Note
Trend	+11.0°F over full period	Clear upward bias; validates warming signal
Seasonal	0.0 (by construction)	Peak-to-trough ≈ 8.7°F; stable across centuries
Residual	≈ 0.0	Low variance (σ≈0.5°F); no major unexplained patterns

The presence of a clean seasonal signal plus a smooth trend supports both data quality and the additive-decomposition approach. Residuals' lack of autocorrelation (checked with ACF) further validates the model choice.

6.4 Discussion and Interpretation

Putting the pieces together: the analysis provides strong, straightforward evidence of long-term, human-amplified warming, with land temperatures acting as an early and clear indicator. The high inter-variable correlations show that various land temperature measures tell the same story — they are redundant but mutually confirming. That synchronicity implies that, for many broad surveillance tasks, land metrics can serve as a practical proxy for larger-scale changes, though ocean processes must still be considered for full climate dynamics.

The rapid post-1980 acceleration aligns temporally with rising greenhouse gas forcings and broader findings in climate science literature (e.g., Hansen and IPCC summaries). Seasonal cycles remain intact — summers are still summers — but the baseline on which they sit keeps lifting, which has implications for extremes (heat waves, stress on ecosystems, etc.). Early-data sparsity (notably before ~1850) is a limitation: fewer stations mean greater uncertainty in the oldest portions of the record. Still, interpolation and robustness checks suggest that the upward trend is not an artifact of data handling.

From an applied perspective, these findings are useful for classroom demonstrations and basic policy communication: clear plots and simple tables make the trend easy to understand. For future work (see Chapter 7), adding spatial breakdowns or coupling with greenhouse gas records would sharpen causal interpretations and reveal regional variability.

Concluding remark

The dataset and analysis together present a coherent narrative: long-term warming, strong internal agreement among land-based measurements, and persistent seasonal cycles. The magnitude and speed of recent warming underscore the urgency of mitigation and monitoring efforts.

CHAPTER 7: CONCLUSION AND FUTURE SCOPE

This chapter brings together the major outcomes of the project and reflects on what the exploratory data analysis has revealed about long-term global temperature behavior. By taking the raw GlobalTemperatures.csv dataset and working through cleaning, visualization, correlation analysis, and seasonal decomposition, the study turned centuries of climate records into understandable patterns and findings. These outcomes tie back directly to the objectives outlined at the beginning of the report and also contribute meaningfully to introductory climate science discussions. The chapter ends by pointing to several ways the work could be expanded or strengthened in the future.

7.1 Conclusion

The project applied a full suite of Python-based EDA techniques to examine **266 years** (1750–2015) of global land temperature data from the Berkeley Earth archive. Before any meaningful analysis could happen, the dataset needed careful preprocessing: converting dates, removing uncertainty columns, handling missing values through interpolation, and creating Fahrenheit-converted variables for easier interpretation. These steps helped clean up the 3,192 monthly entries and addressed gaps in early historical records, setting a solid foundation for consistent trend analysis.

The correlation heatmap (Figure 3) proved especially revealing. It showed extremely strong relationships among temperature variables — for instance, an r-value of **0.98** between land-only and combined land–ocean averages. This level of agreement supports the idea that land temperatures, which respond more quickly to atmospheric changes, often act as the main driver behind global-average temperature signals. This observation is consistent with broader scientific work, including NOAA’s climate assessments.

The seasonal decomposition (Figure 4) added another layer of clarity. By separating the data into trend, seasonal, and residual components, it became obvious how much the long-term warming stands apart from the stable annual cycle. Over the entire period, the trend rose by about **11°F**, and more than half of that increase occurred after 1980 — a striking feature also reflected in the yearly trend plot (Figure 2). Meanwhile, the 12-month seasonal pattern stayed almost unchanged, with predictable July peaks and January dips, and the residuals remained small and non-structured. This reinforces both the reliability of the dataset and the suitability of the additive decomposition model.

Taken together, the project successfully:

- Processed, cleaned, and organized more than two centuries of climate data into usable analytical formats.
- Demonstrated strong internal consistency among land temperature metrics through correlation analysis.
- Decoupled long-term warming from natural seasonality using decomposition techniques.
- Produced clear, publication-ready visualizations using Python libraries such as Pandas, Seaborn, and Statsmodels.

The findings echo the scientific consensus on human-driven global warming. The sharp post-1980 rise — roughly **0.20°F per year**, compared to about **0.01°F per year** before 1900 — aligns with greenhouse gas increases and large-scale land-use changes. Beyond fulfilling the project's aims, the work also highlights how accessible analytical tools can help early-stage engineering students build real evidence-based understanding of climate trends.

7.2 Future Scope

Although the present analysis covers the essential patterns in long-term temperature behavior, there are several natural directions to expand this work — both technically and in terms of real-world usefulness.

1. **Add CO_2 Concentration Comparisons**
   Bringing in historical CO_2 datasets (for example, NOAA's Mauna Loa records) would help examine how temperature rise aligns with atmospheric carbon levels. Plotting temperature against CO_2 concentrations, or running simple regression models, could begin bridging the gap between correlation and causal interpretation.
2. **Build an Interactive Dashboard**
   Using libraries like Streamlit or Plotly Dash, the entire analysis could be turned into a small web app. Users could explore the yearly trends, switch variables on and off, or zoom in and out of specific periods. A feature like this

would be valuable for teaching, presentations, or policy outreach, as it puts the data directly into a viewer's hands.

3. **Extend the Study with Geographic Layers**
   Berkeley Earth provides regional and country-level temperature datasets. Integrating these with tools like GeoPandas or Folium could enable the creation of global and regional climate maps, showing how different parts of the world have warmed over time. This shift from global averages to spatial detail would add depth and strengthen climate justice discussions.

4. **Compare With Paleoclimate Proxy Data**
   Instrumental temperature measurements only reach back so far. Incorporating proxy records — tree rings, ice core isotopes, lake sediments — would help position the observed warming within a much longer historical context. Overlaying proxy estimates with the 1750–2015 dataset would highlight how unusual recent warming truly is.

5. **Develop a Public-Facing Educational Tool**
   Beyond analysis, the project could evolve into an interactive learning platform hosted online. Features like quizzes, guided walkthroughs of decomposition outputs, or simulation sliders (e.g., "What if CO_2 reaches 500 ppm?") would make climate data more approachable for young students or the general public.

These ideas are technically achievable on modest hardware and would transform the project from a mostly descriptive EDA task into a more comprehensive climate-analytics system. Adding predictive models (like basic machine learning or LSTM-based forecasting) could be explored later, though that would shift the project's scope significantly.

Closing Remark

In the end, this project not only highlights the unmistakable upward march of global land temperatures but also equips emerging engineers and data scientists with the tools needed to analyse such trends themselves. As climate pressures intensify worldwide, skills in data interpretation and clear communication will be increasingly valuable — and projects like this help build that foundation.

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